

Geometry Set up

From the geometry of the robotic arm, the end-effector forms a triangle with the origin and the intermediate joint. The parameters are defined as:

Link 1 length: L_1

Link 2 length: L_2

Target coordinates: (x, y)

Distance from origin to target: $r = \sqrt{x^2 + y^2}$

2. Forward Kinematics

The position equations for the end-effector, derived from forward kinematics, are:

$$x = L_1 \cos \theta_1 + L_2 \cos(\theta_1 + \theta_2)$$

$$y = L_1 \sin \theta_1 + L_2 \sin(\theta_1 + \theta_2)$$

We must invert these equations to solve for the unknown joint angles, θ_1 and θ_2 .

3. Solving for θ_2 (Elbow Angle)

To eliminate θ_1 , we square and add the forward kinematics equations:

$$x^2 + y^2 = (L_1 \cos \theta_1 + L_2 \cos(\theta_1 + \theta_2))^2 + (L_1 \sin \theta_1 + L_2 \sin(\theta_1 + \theta_2))^2$$

Expanding the terms and utilizing the trigonometric identity $\cos(A-B) = \cos A \cos B + \sin A \sin B$ simplifies the equation to:

$$x^2 + y^2 = L_1^2 + L_2^2 + 2L_1L_2\cos\theta_2$$

Rearranging the formula to isolate $\cos\theta_2$:

$$\cos\theta_2 = \frac{2L_1(L_2x^2 + y^2 - L_1^2 - L_2^2)}{2L_1L_2}$$

Taking the inverse cosine provides two mathematical solutions:

$$\theta_2 = \pm \cos^{-1}(2(L_1(L_2x^2+y^2-L_1^2-L_2^2))$$

For the Above Configuration (Elbow Up), we we select the positive solution by convention:

$$\theta_2 = +\cos^{-1}(2(L_1(L_2x^2+y^2-L_1^2-L_2^2))$$

4. Solving for θ_1 (Base Angle)

To find θ_1 , we define two auxiliary angles based on the geometry of the setup:

* α : The angle from the positive x-axis ~~to~~ to the target point (x,y) .
 $\alpha = \text{atan2}(y,x)$

β : The interior angle between the first link (L_1) and ~~the~~ the vector connecting the origin to the target point (r) .
 $\beta = \text{atan2}(L_2 \sin \theta_2, L_1 + L_2 \cos \theta_2)$

Using these geometric relationships, the base joint angle is determined by:

$$\theta_1 = \alpha - \beta$$

5. Final Exact Solution (Above Configuration)

$$\theta_2 = \cos^{-1}(2(L_1(L_2x^2+y^2-L_1^2-L_2^2))$$

$$\theta_1 = \text{atan2}(y,x) - \text{atan2}(L_2 \sin \theta_2, L_1 + L_2 \cos \theta_2)$$

Part II: Numerical Application

We now apply the derived formulas to the provided problem

Given Parameters:

$$L_1 = 12 \text{ cm}$$

$$L_2 = 10 \text{ cm}$$

$$x = 15.7 \text{ cm}$$

$$y = 12.6 \text{ cm}$$

First, we determine r^2 (the square of the distance to the target):

$$r^2 = x^2 + y^2$$

$$r^2 = 15.7^2 + 12.6^2 = 246.49 + 158.76 = 405.25$$

Next, we calculate the term $\cos\theta_2$:

$$\cos\theta_2 = \frac{2(12)(10)405.25 - 12^2 - 10^2}{2 \cdot 40 \cdot 405.25 - 144 - 100}$$

$$\cos\theta_2 = \frac{240405.25 - 144 - 100}{240405.25 - 144 - 100}$$

$$\cos\theta_2 = \frac{240161.25}{240161.25} = 0.671875$$

a) Above Configuration (Elbow Up)

For the Above configuration, we take the positive inverse cosine:

$$\theta_2 = \cos^{-1}(0.671875)$$

$$\theta_2 \approx 47.79^\circ$$

To find θ_1 , we calculate α and β :

$$\alpha = \tan^{-1}(12.6, 15.7)$$

$$\alpha \approx 38.75^\circ$$

$$\beta = \tan^{-1}(10 \sin(47.79^\circ), 12 + 10 \cos(47.79^\circ))$$

$$\beta = \tan^{-1}(10 \times 0.7407, 12 + 10 \times 0.6719)$$

$$\beta = \tan^{-1}(7.407, 18.719)$$

$$\beta \approx 21.59^\circ$$

Solving for θ_1 :

$$\theta_1 = \alpha - \beta$$

$$\theta_1 = 38.75^\circ - 21.59^\circ = 17.16^\circ$$

Final Answer Answer for (a):

$$\theta_1 \approx 17.16^\circ$$

$$\theta_2 \approx 47.79^\circ$$

b) Below Configuration (Elbow Down)

For the Below configuration, we use the negative mathematical solution for θ_2 :

$$\theta_2 = -\cos^{-1}(0.671875)$$

$$\theta_2 \approx -47.79^\circ$$

We use the same formula for θ_1 , substituting the negative θ_2 value. The angle α remains 38.75° .

$$\beta_{\text{below}} = \text{atan2}(10 \sin(-47.79^\circ), 12 + 10 \cos(-47.79^\circ))$$

$$\beta_{\text{below}} = \text{atan2}(-7.407, 18.719)$$

$$\beta_{\text{below}} \approx -21.59^\circ$$

Solving for the new θ_1 :

$$\theta_1 = \alpha - \beta_{\text{below}}$$

$$\theta_1 = 38.75^\circ - (-21.59^\circ)$$

$$\theta_1 = 38.75^\circ + 21.59^\circ = 60.34^\circ$$

Final Answer for (b):

$$\theta_1 \approx 60.34^\circ$$

$$\theta_2 \approx -47.79^\circ$$